Celanese
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Zytel® FR50 BK505

NYLON RESIN

Common features of Zytel® nylon resin include mechanical and physical properties such as high mechanical strength, excellent balance of stiffness and toughness, good high temperature performance, good electrical and flammability properties, good abrasion and chemical resistance. In addition, Zytel® nylon resins are available in different modified and reinforced grades to create a wide range of products with tailored properties for specific processes and end-uses. Zytel® nylon resin, including most flame retardant grades, offer the ability to be coloured.

The good melt stability of Zytel® nylon resin normally enables the recycling of properly handled production waste. If recycling is not possible, we recommend, as the preferred option, incineration with energy recovery (-31kJ/g of base polymer) in appropriately equipped installations. For disposal, local regulations have to be observed.

Zytel® nylon resin typically is used in demanding applications in the automotive, furniture, domestic appliances, sporting goods and construction industry.

Zytel® FR50 BK505 is a 25% glass fiber reinforced, flame retardant polyamide 66 resin for injection molding.

Product information

Resin Identification	PA66-GF25 FR(17)	ISO 1043
Part Marking Code	>PA66-GF25 FR(17)<	ISO 11469
ISO designation	ISO 16396-PA66,GF25 FR(17),M1CF1GR,S14-110	
Infrared spectrum	available	

Rheological properties

	dry/cond.		
Viscosity number	150[1]/*[DS]	cm^3/g	ISO 307, 1628
Molding shrinkage, parallel	0.3 / -	%	ISO 294-4, 2577
Molding shrinkage, normal	0.7 / -	%	ISO 294-4, 2577

[DS]: Derived from similar grade

[1]: Sulfuric acid 96%

Typical mechanical properties

	dry/cond.		
Tensile modulus	11000 / 8000	MPa	ISO 527-1/-2
Tensile stress at break, 5mm/min	160 / 120	MPa	ISO 527-1/-2
Tensile strain at break, 5mm/min	2.1 / 3	%	ISO 527-1/-2
Flexural modulus	9500 / 7500	MPa	ISO 178
Charpy impact strength, 23 °C	65 / -	kJ/m^2	ISO 179/1eU
Charpy impact strength, -30 °C	60 / -	kJ/m^2	ISO 179/1eU
Charpy impact strength, -40 °C	50 / -[DS]	kJ/m^2	ISO 179/1eU
Charpy notched impact strength, 23 °C	10 / 12	kJ/m^2	ISO 179/1eA
Charpy notched impact strength, -30 °C	9 / 11	kJ/m^2	ISO 179/1eA
Charpy notched impact strength, -40 °C	9 / 10	kJ/m^2	ISO 179/1eA

[DS]: Derived from similar grade

Thermal properties

	dry/cond.		
Melting temperature, 10 °C/min	260[2]/*	°C	ISO 11357-1/-3
Glass transition temperature, 10 °C/min	80 / 20	°C	ISO 11357-1/-3
Temperature of deflection under load, 1.8 MPa	242 / *	°C	ISO 75-1/-2
Coefficient of linear thermal expansion (CLTE), parallel	21 / *	E-6/K	ISO 11359-1/-2

Zytel® FR50 BK505

NYLON RESIN

Coefficient of linear thermal expansion (CLTE), parallel, -40-23°C	25 / *	E-6/K	ISO 11359-1/-2
Coefficient of linear thermal expansion (CLTE), parallel, 55-160°C	12 / *	E-6/K	ISO 11359-1/-2
Coefficient of linear thermal expansion (CLTE), normal	79 / *	E-6/K	ISO 11359-1/-2
Coefficient of linear thermal expansion (CLTE), normal, -40-23°C	69 / *	E-6/K	ISO 11359-1/-2
Coefficient of linear thermal expansion (CLTE), normal, 55-160°C	110 / *	E-6/K	ISO 11359-1/-2
Thermal conductivity of melt	0.25	W/(m K)	ISO 22007-2
Specific heat capacity of melt	2000	J/(kg K)	ISO 22007-4
RTI, electrical, 0.75mm	130	°C	UL 746B
RTI, electrical, 1.5mm	130	°C	UL 746B
RTI, electrical, 3.0mm	130	°C	UL 746B
RTI, impact, 0.75mm	105	°C	UL 746B
RTI, impact, 1.5mm	115	°C	UL 746B
RTI, impact, 3.0mm	115	°C	UL 746B
RTI, strength, 0.75mm	105	°C	UL 746B
RTI, strength, 1.5mm	115 / *	°C	UL 746B
RTI, strength, 3.0mm	120	°C	UL 746B

[2]: 1st heating

Flammability

dry/cond.

Burning Behav. at 1.5mm nom. thickn.	V-0 / *	class	IEC 60695-11-10
Thickness tested	1.5 / *	mm	IEC 60695-11-10
UL recognition	yes / *		UL 94
Burning Behav. at thickness h	V-0 / *	class	IEC 60695-11-10
Thickness tested	0.35 / *	mm	IEC 60695-11-10
UL recognition	yes[3]/*		UL 94
Burning Behav. 5V at thickness h	5VA / *	class	IEC 60695-11-20
Thickness tested	1.5 / *	mm	IEC 60695-11-20
UL recognition	yes / *		UL 94
Oxygen index	35 / *	%	ISO 4589-1/-2
Glow Wire Flammability Index, 0.75mm	960 / -	°C	IEC 60695-2-12
Glow Wire Flammability Index, 1.5mm	960 / -	°C	IEC 60695-2-12
Glow Wire Flammability Index, 3.0mm	960 / -	°C	IEC 60695-2-12
Glow Wire Ignition Temperature, 0.75mm	900 / -	°C	IEC 60695-2-13
Glow Wire Ignition Temperature, 1.5mm	900 / -	°C	IEC 60695-2-13
Glow Wire Ignition Temperature, 3.0mm	930 / -	°C	IEC 60695-2-13
FMVSS Class	SE / B		ISO 3795 (FMVSS 302)

[3]: f1

Electrical properties

dry/cond.

Volume resistivity	>1E13 / 2.7E10[DS]	Ohm.m	IEC 62631-3-1
Comparative tracking index	275 / -[DS]		IEC 60112
Electric Strength, Short Time, 1mm	30 / -	kV/mm	IEC 60243-1

Electric Strength, Short Time, 2mm	24 / 22[DS]	kV/mm	IEC 60243-1

[DS]: Derived from similar grade

Physical/Other properties

	dry/cond.		
Humidity absorption, 2mm	1.3 / *	%	Sim. to ISO 62
Water absorption, 2mm	3.4 / *[DS]	%	Sim. to ISO 62
Water absorption, Immersion 24h	0.6[4]/*	%	Sim. to ISO 62
Density	1600 / -	kg/m^3	ISO 1183

[DS]: Derived from similar grade

[4]: thickness,2mm

Emissions

Emission of organic compounds	4.7	µgC/g	VDA 277
Odor test	4.5	class	VDA 270

Injection

Drying Recommended	yes	
Drying Temperature	80	°C
Drying Time, Dehumidified Dryer	2 - 4	h
Processing Moisture Content	≤0.2	%
Melt Temperature Optimum	290	°C
Min. melt temperature	280	°C
Max. melt temperature	300	°C
Max. screw tangential speed	≤0.2	m/s
Mold Temperature Optimum	100	°C
Min. mold temperature	70	°C
Max. mold temperature	120	°C
Hold pressure range	50 - 100	MPa
Hold pressure time	3	s/mm
Hold Pressure Time (h is the max. wall thickness of the part in mm)	h^2+2	s
Ejection temperature	210	°C

Automotive

OEM	STANDARD	ADDITIONAL INFORMATION
CATL	MCR0000561	
General Motors	GMW17481P-PA66-GF25	Black
Hyundai	MS941-03 Type A-5 FRV0	
Stellantis	B62 0300 / 61/223E-216M/C1/C4	01378_19_02645

Characteristics

Processing	Injection Molding
Additives	Flame retardant
Special characteristics	Flame retardant

Zytel® FR50 BK505

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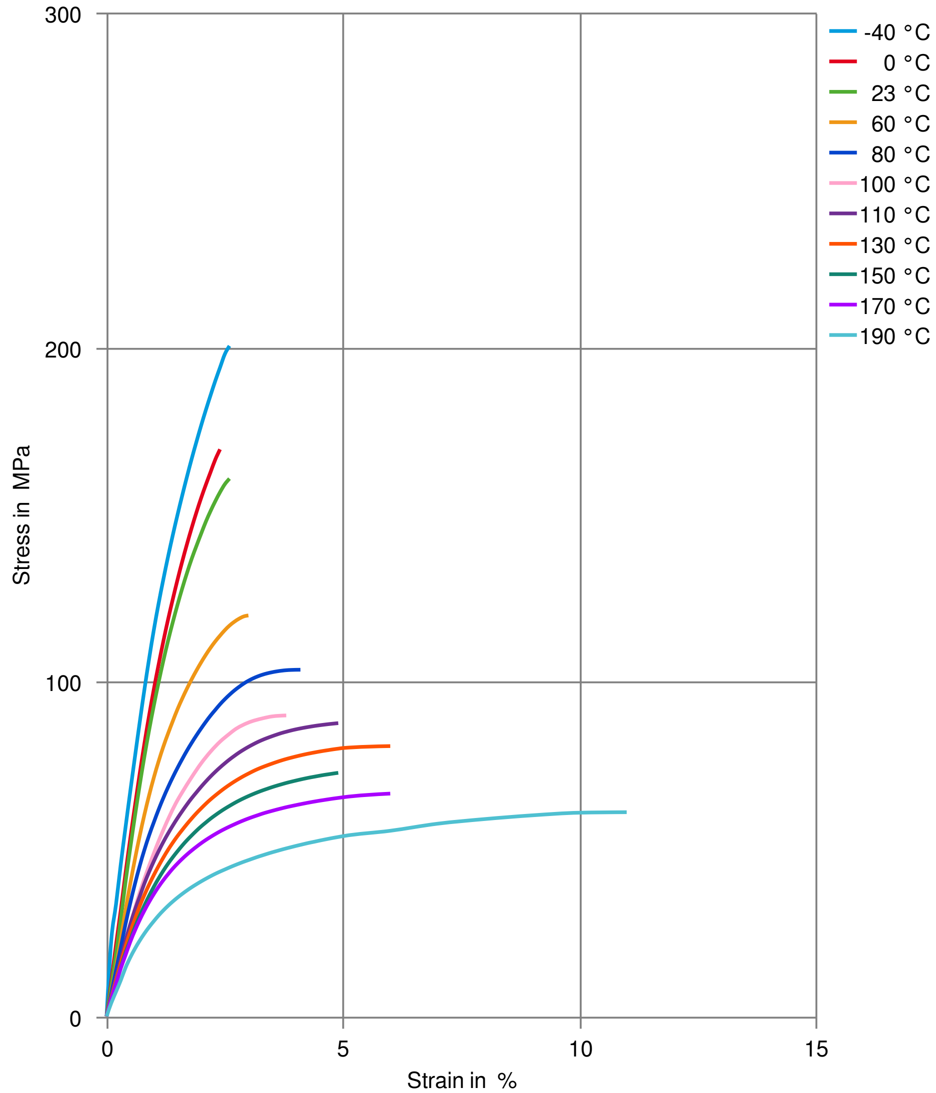
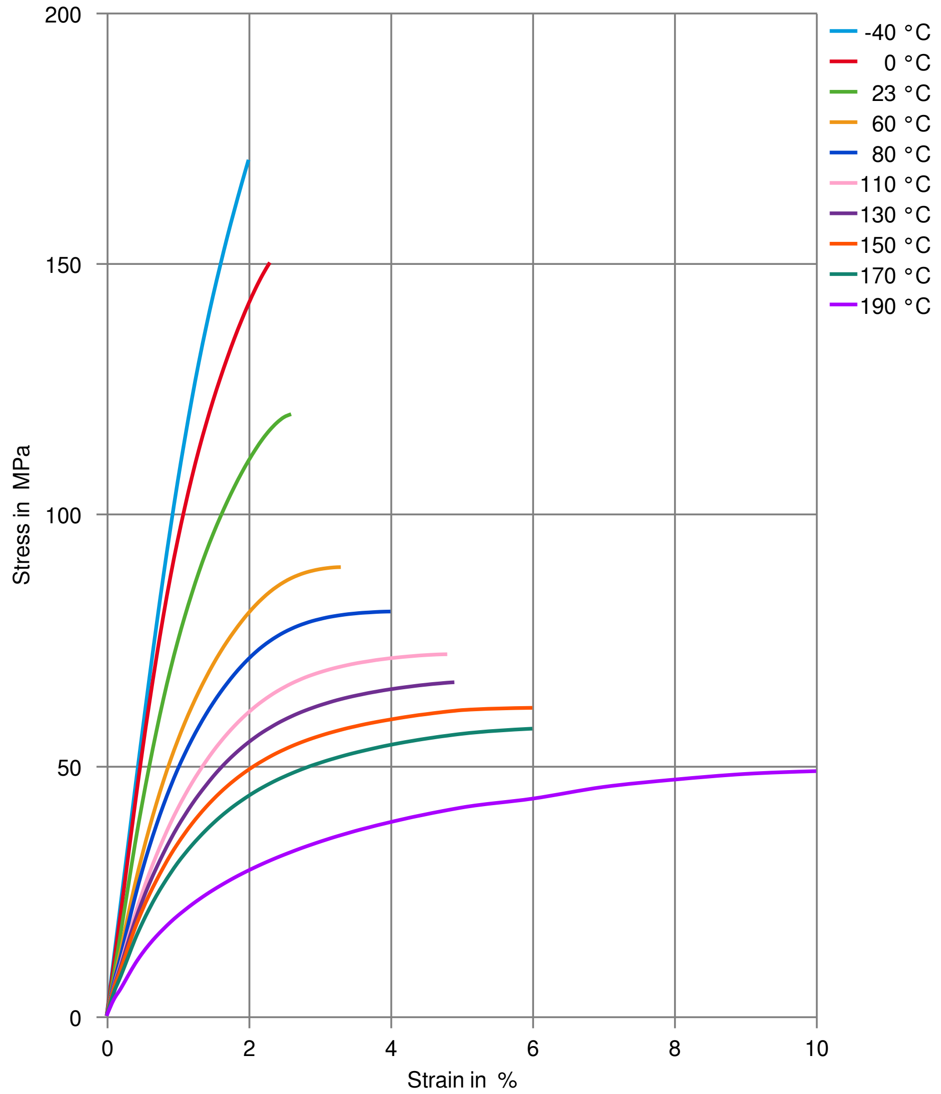
Stress-strain (dry)



Zytel® FR50 BK505

NYLON RESIN

Stress-strain (cond.)



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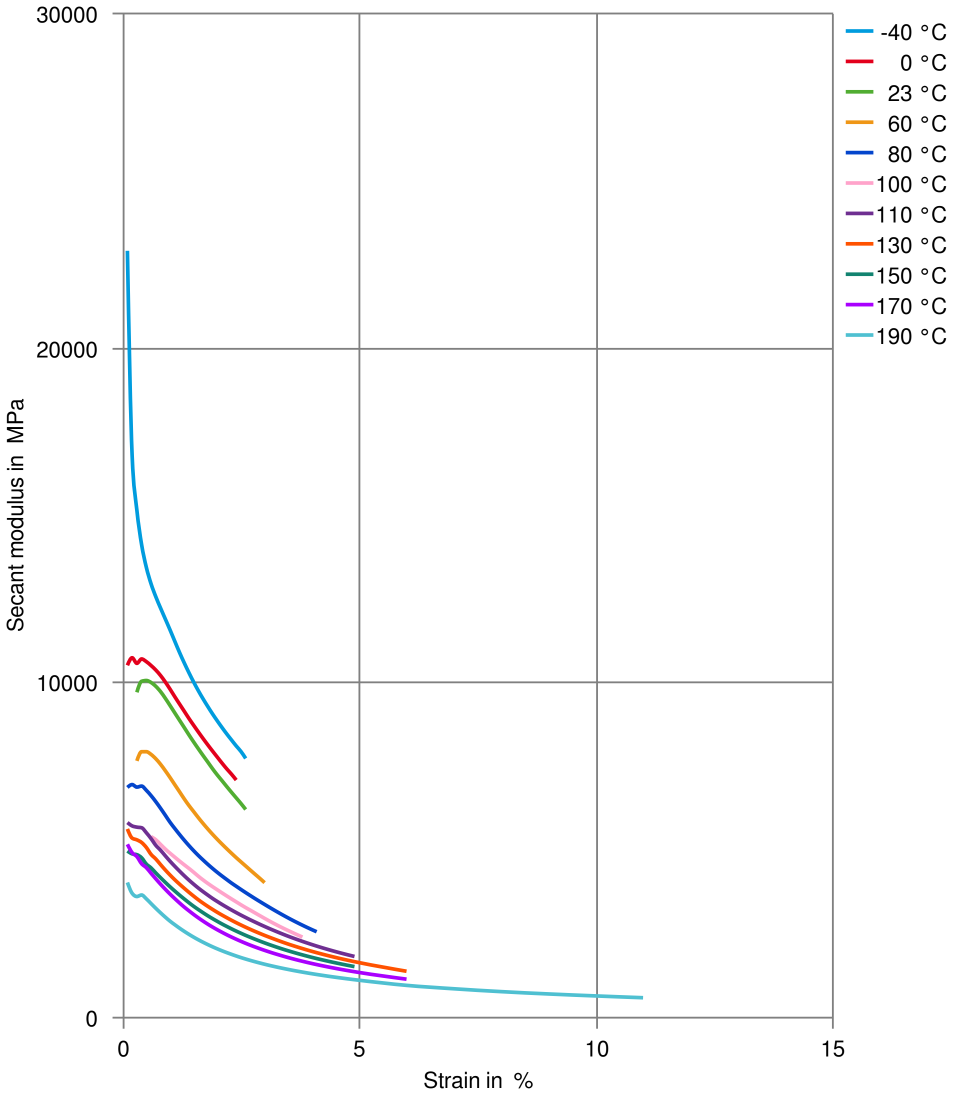
Zytel® FR50 BK505

NYLON RESIN

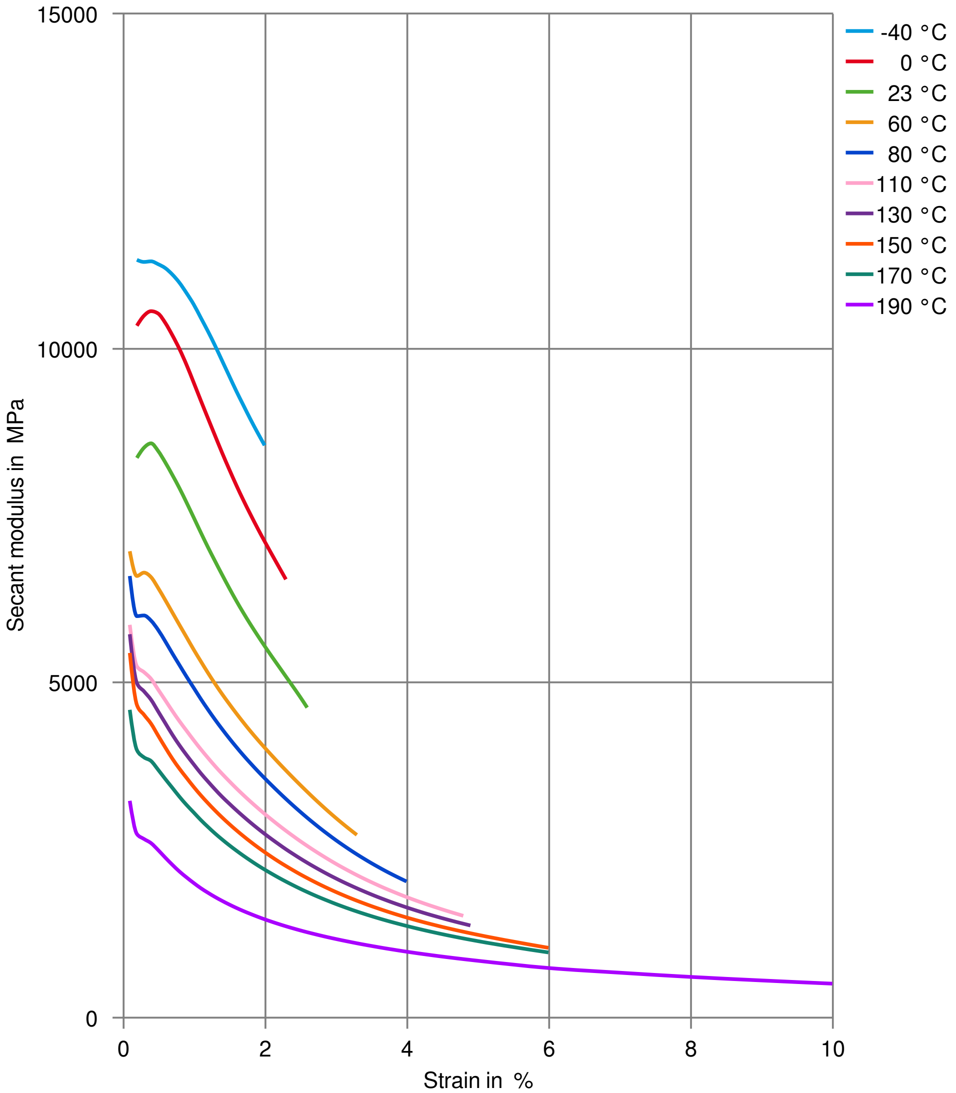
Secant modulus-strain (dry)



Zytel® FR50 BK505

NYLON RESIN

Secant modulus-strain (cond.)



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Chemical Media Resistance

Acids

- ✔ Acetic Acid (5% by mass), 23°C
- ✔ Citric Acid solution (10% by mass), 23°C
- ✔ Lactic Acid (10% by mass), 23°C
- ✖ Hydrochloric Acid (36% by mass), 23°C
- ✖ Nitric Acid (40% by mass), 23°C
- ✖ Sulfuric Acid (38% by mass), 23°C
- ✖ Sulfuric Acid (5% by mass), 23°C
- ✖ Chromic Acid solution (40% by mass), 23°C

Bases

- ✖ Sodium Hydroxide solution (35% by mass), 23°C
- ✔ Sodium Hydroxide solution (1% by mass), 23°C
- ✔ Ammonium Hydroxide solution (10% by mass), 23°C

Alcohols

- ✔ Isopropyl alcohol, 23°C
- ✔ Methanol, 23°C
- ✔ Ethanol, 23°C

Hydrocarbons

- ✔ n-Hexane, 23°C
- ✔ Toluene, 23°C
- ✔ iso-Octane, 23°C

Ketones

- ✔ Acetone, 23°C

Ethers

- ✔ Diethyl ether, 23°C

Mineral oils

- ✔ SAE 10W40 multigrade motor oil, 23°C
- ✔ SAE 10W40 multigrade motor oil, 130°C
- ✔ SAE 80/90 hypoid-gear oil, 130°C
- ✔ Insulating Oil, 23°C

Standard Fuels

- ✔ ISO 1817 Liquid 1 - E5, 60°C
- ✔ ISO 1817 Liquid 2 - M15E4, 60°C
- ✔ ISO 1817 Liquid 3 - M3E7, 60°C
- ✔ ISO 1817 Liquid 4 - M15, 60°C
- ✔ Standard fuel without alcohol (pref. ISO 1817 Liquid C), 23°C
- ✔ Standard fuel with alcohol (pref. ISO 1817 Liquid 4), 23°C
- ✔ Diesel fuel (pref. ISO 1817 Liquid F), 23°C
- ✔ Diesel fuel (pref. ISO 1817 Liquid F), 90°C
- ✔ Diesel fuel (pref. ISO 1817 Liquid F), >90°C

Salt solutions

- ✔ Sodium Chloride solution (10% by mass), 23°C
- ✖ Sodium Hypochlorite solution (10% by mass), 23°C

Zytel® FR50 BK505

NYLON RESIN

- ✔ Sodium Carbonate solution (20% by mass), 23 °C
- ✔ Sodium Carbonate solution (2% by mass), 23 °C
- ✘ Zinc Chloride solution (50% by mass), 23 °C

Other

- ✔ Ethyl Acetate, 23 °C
- ✘ Hydrogen peroxide, 23 °C
- ✔ DOT No. 4 Brake fluid, 130 °C
- ✔ Ethylene Glycol (50% by mass) in water, 108 °C
- ✔ 1% nonylphenoxy-polyethyleneoxy ethanol in water, 23 °C
- ✔ 50% Oleic acid + 50% Olive Oil, 23 °C
- ✔ Water, 23 °C
- ✘ Water, 90 °C
- ✘ Phenol solution (5% by mass), 23 °C

Symbols used:

✔ possibly resistant
Defined as: Supplier has sufficient indication that contact with chemical can be potentially accepted under the intended use conditions and expected service life. Criteria for assessment have to be indicated (e.g. surface aspect, volume change, property change).

✘ not recommended - see explanation
Defined as: Not recommended for general use. However, short-term exposure under certain restricted conditions could be acceptable (e.g. fast cleaning with thorough rinsing, spills, wiping, vapor exposure).

Revised: 2025-07-08 Source: Celanese Materials Database